TRIBHUVAN UNIVERSITY — BBA 8TH SEMESTER

BNK 209: INSURANCE AND RISK MANAGEMENT — COMPLETE EXAM REVISION NOTES
Unit-by-Unit Master Guide (Units 1 to 7) | Legal Principles, Actuarial Numericals, Nepal NIA Laws & Case Guide

UNIT 1: INTRODUCTION TO RISK (5 LHs)

1.1 Core Definitions: Risk, Uncertainty, Peril, Hazard & Loss Exposure

• **Risk vs. Uncertainty:**

- **Risk:** Uncertainty concerning the occurrence of a financial loss where the **probabilities of possible outcomes can be objectively measured or estimated** from historical data (objective, measurable, insurable).
- **Uncertainty:** A subjective psychological state of doubt about the future where **probabilities cannot be objectively measured** (subjective, unmeasurable).
- **Objective Risk (Degree of Risk):** Relative variation of actual loss from expected loss: Objective Risk = Standard Deviation / Expected Loss (decreases as the number of exposure units *N* increases by the Law of Large Numbers).

• **Peril:** The direct, immediate cause of a loss (e.g., fire, lightning, earthquake, flood, auto collision, burglary).

• **Hazard:** Any condition that creates or increases the frequency or severity of a loss arising from a given peril:

1. **Physical Hazard:** Physical condition of the asset/environment (e.g., icy roads, defective electrical wiring, storing petrol near a furnace).
2. **Moral Hazard:** Dishonesty or character defects in an individual that induce intentional loss or fraud (e.g., arson to collect fire insurance, faking an accident).
3. **Attitudinal (Morale) Hazard:** Carelessness or indifference to a loss because one has insurance ("It's insured, so why lock the car?").
4. **Legal Hazard:** Characteristics of the legal/court system or regulatory environment that increase the frequency or size of liability awards.

1.2 Classification of Risk (Major Exam Differentiations)

Risk Pair	First Category	Second Category
1. Pure vs. Speculative Risk	**Pure Risk:** Only two outcomes—**Loss or No Loss** (no gain). Accidental, involuntary, and **insurable** (e.g., fire, earthquake, premature death).	**Speculative Risk:** Three outcomes—**Loss, No Loss, or Profit/Gain**. Voluntary, entrepreneurial, and **not insurable** (handled via hedging/diversification; e.g., stock trading, launching a startup).
2. Diversifiable vs. Non-Diversifiable	**Diversifiable (Particular / Nonsystematic) Risk:** Affects only individuals or a small group; can be reduced by diversification/pooling (e.g., factory fire, auto theft).	**Non-Diversifiable (Fundamental / Systemic) Risk:** Affects the entire economy or large population simultaneously; cannot be eliminated by simple pooling (e.g., war, hyperinflation, pandemic, systemic financial collapse).
3. Enterprise vs. Systemic Risk	**Enterprise Risk:** Encompasses all risks of a single firm—pure, financial/speculative, operational, strategic, and reputational—managed under ERM.	**Systemic Risk:** Risk of collapse of an entire financial system or market due to interlinkages, where one major bank failure triggers a domino cascade.

1.3 Major Personal Risks, Commercial Loss Exposures & Burden of Risk on Society

• **Major Personal Risks (Affecting Individuals & Families):**
(1) **Premature Death Risk** (death of breadwinner leaving unfulfilled obligations), (2) **Inadequate Retirement Income / Old-Age Risk** (living too long/outliving savings), (3) **Poor Health Risk** (catastrophic medical bills + disability loss of earned income), (4) **Unemployment Risk**, and (5) **Personal Property & Liability Risk**.

• **Major Commercial Risks (Loss Exposures of Businesses):**
(1) **Property Loss Exposure** (Direct damage to building/servers/inventory + Indirect Business Interruption loss of profit), (2) **Liability Loss Exposure** (Premises, Products, Motor Third-Party, Professional E&O;, Employers' liability), (3) **Human Resource / Key-Person Exposure**, and (4) **Cyber Security, Data Breach & Crime/Fidelity Exposure**.

• **Burden of Risk on Society (4 Key Costs):**
(1) *Requirement of Larger Emergency Reserves* (ties up idle capital), (2) *Deprivation of Socially Useful Goods & Services*

(firms avoid high-risk innovations), (3) *Mental Fear, Worry & Anxiety*, and (4) *Direct/Indirect Cost of Uncompensated Disasters*.

UNIT 2: RISK MANAGEMENT (6 LHs)

2.1 Concept, Objectives & 6-Step Process of Risk Management

• **Meaning of Risk Management:** A systematic scientific process of identifying, analyzing, and evaluating loss exposures faced by an organization or individual, and selecting and monitoring the most appropriate combination of risk control and risk financing techniques.

• **Objectives of Risk Management:**

- **Pre-Loss Objectives:** (1) Economy (least cost preparation), (2) Reduction of anxiety/worry, (3) Meeting legal/statutory obligations.

- **Post-Loss Objectives:** (1) Survival of the firm, (2) Continuity of operations, (3) Stability of earnings, (4) Continued growth, (5) Social responsibility.

• **6-Step Process of Risk Management:**

Step 1: Determine Objectives -> **Step 2:** Identify Loss Exposures -> **Step 3:** Measure & Analyze Loss Exposures (Frequency & Severity / Loss Forecasting) -> **Step 4:** Select Risk Treatment Techniques (Risk Control + Risk Financing) -> **Step 5:** Implement the Program -> **Step 6:** Monitor & Review Regularly (via RMIS).

Risk Treatment Matrix (Frequency vs. Severity of Loss)

1. LOW Frequency + LOW Severity **Best Technique: RETENTION (Self-Insure)** (Absorb out of working capital / small deductible)	**2. HIGH Frequency + LOW Severity** **Best Technique: LOSS PREVENTION + RETENTION** (Predictable routine losses; use safety + budget)
3. LOW Frequency + HIGH Severity **Best Technique: COMMERCIAL INSURANCE** (Transfer catastrophic risk to an insurer!)	**4. HIGH Frequency + HIGH Severity** **Best Technique: AVOIDANCE (+ Loss Reduction)** (Too expensive to insure; abandon the activity!)

2.2 Risk Control vs. Risk Financing, Loss Forecasting, RMIS, VaR & ERM

• **1. Risk Control Techniques (Reduce Frequency or Severity):**

- *Avoidance:* Refusing to engage in the risky activity (0% frequency).

- *Loss Prevention:* Reduces **loss frequency** (driver training, firewall, safety rules).

- *Loss Reduction:* Reduces **loss severity** after a peril occurs (fire sprinklers, airbags, off-site data duplication, warehouse separation).

• **2. Risk Financing Techniques (Pay for Losses That Occur):**

- *Retention:* Active or passive self-insurance via working capital, deductibles, funded reserves, or a **Captive Insurer**.

- *Non-Insurance Transfer:* Hold-harmless contracts, leases, outsourcing, or derivative hedging.

- *Commercial Insurance:* Transferring low-frequency, high-severity risks to an insurer.

• **3. Loss Forecasting (3 Quantitative Techniques):**

(1) **Probability Analysis** (Expected Loss = Frequency x Severity), (2) **Regression Analysis** (predicting claims Y from physical drivers X like payroll or fleet mileage), and (3) **Forecasting Based on Loss Distributions** (Poisson for frequency; Lognormal/Pareto for severity).

• **4. Risk Appetite vs. Risk Tolerance & Value at Risk (VaR):**

- *Risk Appetite:* Broad strategic level of risk a firm is willing to accept to pursue its goals.

- *Risk Tolerance:* Specific, quantitative maximum acceptable variance around a specific target.

- *Value at Risk (VaR):* Maximum expected rupee loss over a given time period (e.g., 10 days) at a specified confidence level (e.g., 99%).

- *Risk Management Information System (RMIS):* Computerized database tracking exposures, policies, claims, and loss trends.

UNIT 3: THE INSURANCE DEVICE (5 LHs)

3.1 Definition, Basic Characteristics & Ideally Insurable Risk

• **Four Basic Characteristics of Insurance:**
(1) **Pooling of Losses** (spreading losses of the few over the entire group so average loss substitutes for actual loss), (2) **Payment of Fortuitous (Accidental) Losses**, (3) **Risk Transfer** (from insured to insurer), and (4) **Indemnification** (restoring pre-loss financial position).
• **Six Requirements of an Ideally Insurable Risk (Why Catastrophes Are Hard to Insure):**
1. **Large Number of Homogeneous Exposure Units** (so the Law of Large Numbers works).
2. **Accidental and Unintentional Loss** (fortuitous; outside the insured's control).
3. **Determinable and Measurable Loss** (clear cause, time, place, and Rupee amount).
4. **No Catastrophic Loss** (exposures must be independent; earthquakes/wars violate this because thousands of units suffer loss simultaneously).
5. **Calculable Chance of Loss** (actuaries can estimate average frequency and severity).
6. **Economically Feasible Premium** (premium must be affordable relative to the sum insured).

3.2 Law of Large Numbers, Adverse Selection, Insurance vs. Gambling & Hedging

• **Law of Large Numbers in Risk Pooling:** States that as the number of independent, homogeneous exposure units in a pool increases, the actual observed average loss approaches the true expected probability loss—reducing the insurer's objective risk and enabling accurate rate-making.

• **Adverse Selection (Anti-Selection) & 5 Control Tools:**

- *Definition:* Tendency of higher-than-average risk applicants to seek insurance at standard rates more aggressively than low-risk people (due to asymmetric information).

- *5 Control Tools:* (1) Strict **Underwriting & Risk Classification** (preferred, standard, substandard, declined), (2) **Waiting Periods & Pre-Existing Condition Exclusions**, (3) **Deductibles, Coinsurance & Policy Limits**, (4) **Group Underwriting**, and (5) **Enforcing Utmost Good Faith**.

• **Insurance vs. Gambling:** Insurance transfers an *existing pure risk*, requires *insurable interest*, follows *indemnity*, and is socially productive; Gambling creates a *new speculative risk*, requires no insurable interest, and is socially unproductive.

• **Insurance vs. Hedging:** Insurance transfers *insurable pure risk* via a pooling contract (reducing objective risk via the Law of Large Numbers); Hedging transfers *speculative price/market risk* (commodities, FX, interest rates) via derivatives (forwards, futures, options) to speculators.

UNIT 4: LEGAL PRINCIPLES IN INSURANCE (5 LHs)

4.1 The Six Fundamental Legal Principles of Insurance Contracts (10-Mark Must Know)

Legal Principle	Core Meaning & Legal Rules	Purpose / Practical Application
1. Utmost Good Faith (Uberrimae Fidei)	Higher standard of honesty requiring voluntary disclosure of all material facts. Supported by 3 doctrines: • **Representations** (applicant statements) • **Concealment** (hiding material facts) • **Warranties** (strict policy promises).	Eliminates information asymmetry and prevents fraudulent underwriting.
2. Insurable Interest	The insured must stand in a legal financial relationship to the subject matter such that they suffer a direct pecuniary loss if it is damaged/dies. • **Life Insurance:** Must exist at **inception**. • **Property/Non-Life:** Must exist at **time of loss**.	Prevents wagering/gambling, reduces moral hazard (murder/arson), and measures loss.
3. Principle of Indemnity	Insurer pays no more than the actual financial loss suffered (measured by Actual Cash Value = Replacement Cost - Depreciation). *(Exception: Life & PA insurance are valued contracts).*	Prevents insured from profiting from a loss and reduces moral hazard.
4. Principle of Subrogation	After indemnifying the insured, the insurer steps into the legal shoes of the insured and acquires the right to sue the negligent third party who caused the loss.	Prevents insured from collecting twice for the same loss and holds negligent parties liable.
5. Principle of Contribution	When double insurance exists (two+ policies covering the same property and peril), each insurer pays only its pro-rata share: **Share = Loss x (Policy Limit / Total Limits)**.	Preserves indemnity so total payout across all insurers never exceeds actual loss.

Legal Principle	Core Meaning & Legal Rules	Purpose / Practical Application
6. Proximate Cause (Causa Proxima)	The active, efficient cause that sets in motion an unbroken chain of events leading to the loss must be an **insured peril** (not an excluded peril).	Determines whether complex multi-cause claims are covered.

4.2 Distinct Legal Characteristics, Basic Parts & Policy Provisions

• **Distinct Legal Characteristics of Insurance Contracts:**

1. **Aleatory Contract:** Dollar values exchanged are unequal depending on chance (a small premium can trigger a huge claim payout if a loss occurs).
2. **Unilateral Contract:** Only the insurer makes a legally enforceable promise after the insured pays the upfront premium.
3. **Contract of Adhesion:** Drafted entirely by the insurer on a "take-it-or-leave-it" basis; therefore, any ambiguity in wording is interpreted by courts in favor of the insured.
4. **Conditional Contract:** Insurer's obligation to pay is conditional upon the insured complying with policy conditions (timely notice, proof of loss).
5. **Personal Contract:** In property insurance, the contract insures the person's financial interest (cannot be assigned to a new buyer without insurer consent).

• **Four Basic Parts of an Insurance Contract (D-I-C-E):**

(1) Declarations (who, what, where, policy limits, premium), **(2) Insuring Agreement** (broad coverage promise: *Named-Perils* vs. *Open-Perils*), **(3) Exclusions** (excluded perils, losses, and property), and **(4) Conditions** (duties of insured after loss).

• **Key Policy Provisions:**

- **Endorsements vs. Riders:** Written amendments modifying the policy—called *Endorsements* in Property/Casualty insurance and *Riders* in Life/Health insurance.
- **Deductibles:** Initial amount of loss borne by the insured before the insurer pays (eliminates small nuisance claims and curbs moral/attitudinal hazard).
- **Coinsurance (Condition of Average in Property Insurance):** Requires property to be insured up to a stated percentage (e.g., 80% or 100%) of its true value:

Claim Payout = Actual Loss x (Amount of Insurance Carried / Amount of Insurance Required).

EXAM MEMORY TIP / FORMULA HINT: *Remember the 4 Basic Parts of an Insurance Policy by 'D-I-C-E': Declarations, Insuring Agreement, Conditions, and Exclusions!*

UNIT 5: LIFE INSURANCE & ACTUARIAL NUMERICALS (10 LHs)

5.1 Premature Death, Types of Life Insurance, Riders & Contractual Provisions

• **Premature Death & Financial Impact:** Death of a family head with unfulfilled financial obligations (dependents, unpaid home/car loans, children's education). Causes loss of Human Life Value, funeral/medical clearance costs, debt foreclosure risk, and drop in family living standards.

• **Term Life Insurance vs. Cash-Value Life Insurance (Whole Life / Endowment / Universal Life):**

- **Term Life:** Temporary pure death protection (e.g., 1, 10, 20, 25 yrs); **no cash value / no savings element**; lowest premium; ideal for young breadwinners with high mortgages ("Buy Term and Invest the Difference").
- **Whole Life / Endowment (Cash-Value):** Long-term/permanent protection bundled with a **savings component (Cash Surrender Value + Bonus)**; allows policy loans and surrender; higher level premium.

• **Essential Life Insurance Riders & Provisions:**

- **Riders:** (1) *Accidental Death & Dismemberment (AD&D;) / Permanent Total Disability (PTD) Rider*, (2) *Critical Illness (CI) Rider*, (3) *Waiver of Premium Rider*.
- **Nonforfeiture Options (for Cash-Value Policies):** (1) Cash Surrender Value, (2) Reduced Paid-Up Insurance, (3) Extended Term Insurance.
- **Settlement Options:** Lump-sum cash, Interest option, Fixed-period installments, Fixed-amount installments, or Life income annuity.

• **Taxation of Life Insurance in Nepal (Income Tax Act 2058):**

- **Tax Deduction on Premium:** Resident individuals can deduct annual life insurance premiums paid up to **Rs 40,000** from taxable income (plus Rs 20,000 for health insurance and Rs 5,000 for private house insurance).

- **Payout Tax:** Death benefit payouts are **100% tax-exempt**; maturity investment gains on policies carry a **5% final withholding tax**.

5.2 Step-by-Step Numerical Formulas & Worked Examples for Unit 5 (Exam Favourites)

FORMULA 1: Human Life Value (HLV) Approach:

• **Step 1:** Remaining Working Years (n) = Planned Retirement Age - Current Age

• **Step 2:** Net Annual Family Share = Gross Annual Earnings - (Income Taxes + Personal Self-Maintenance Expenses + Personal Insurance Premiums)

• **Step 3: HLV = Net Annual Family Share x PVIFA(r%, n years)** where PVIFA(r, n) = [1 - (1 + r)^(-n)] / r

Worked Example: Age = 40, Retire = 60 (n = 20 yrs), Gross Income = Rs 600,000, Taxes & Personal Needs = Rs 30,000, r = 6% (PVIFA = 11.4699):

Family Share = 600,000 - 30,000 = Rs 570,000 => **HLV = 570,000 x 11.4699 = Rs 6,537,843**.

FORMULA 2: Financial Needs Approach:

• **Total Life Insurance Needed = (Immediate Cash/Clearance Fund + Mortgage/Loan Payoff Fund + Emergency Fund + Education Fund [PV or Nominal] + PV of Ongoing Family Living Expenses) - Existing Liquid Financial Assets**.

FORMULA 3: Actuarial Life Insurance Premium Calculation (Equivalence Principle):

• **Three Pillars of Life Rate Making:** (1) Mortality Rate (q_x from Mortality Table), (2) Assumed Interest Rate (i), and (3) Expense & Contingency Loading.

• **1-Year Term Net Single Premium (NSP):**

NSP = [Face Value of Policy x Probability of Death (q_x)] / (1 + i)^1

• **Gross (Office) Premium = Net Single Premium (NSP) + Expense Loading**

Worked Example: Age = 35, Policy Face Value = Rs 1,000,000, Probability of death q_35 = 0.0020 (20 deaths per 10,000 lives), Interest rate i = 7%:

Expected Claim = 1,000,000 x 0.0020 = Rs 2,000 => **NSP = Rs 2,000 / 1.07 = Rs 1,869.16**.

EXAM MEMORY TIP / FORMULA HINT: *Never forget the 3 formulas of Unit 5: (1) HLV = (Income - Personal/Tax Cost) x PVIFA(r, n); (2) Needs Approach = Debt Payoff + Emergency + Education + PV(Family Living Expenses) - Existing Assets; (3) 1-Yr Term NSP = (Face Value x q_x) / (1 + i)!*

UNIT 6 & UNIT 7: NON-LIFE INSURANCE & GOVERNMENT REGULATION IN NEPAL (17 LHs)

6.1 Fire Insurance, Marine Insurance, Auto (Motor) Insurance, Agriculture & Social Insurance

• **1. Fire Insurance:**

- *Legal Meaning of "Fire":* Must involve (i) actual ignition/flame and (ii) be accidental/fortuitous and a **"Hostile Fire"** (escaping its normal container).
- *Factors Affecting Fire Rates:* Construction class (RCC vs. wood), Occupancy (office vs. chemical factory), Location/Exposure hazard, Internal fire-fighting appliances (sprinklers/extinguishers), Public fire brigade access, and Past loss history.
- *4 Rate-Making Methods:* (1) **Judgment Rating** (case-by-case), (2) **Class / Manual Tariff Rating** (homogeneous groups), (3) **Schedule Rating** (Basis rate + debits for hazards - credits for fire protection), and (4) **Experience Rating**.

• **2. Marine Insurance (Lifeblood of International Trade):**

- Covers **Marine Hull**, **Marine Cargo** (Institute Cargo Clauses A, B, C), and **Freight**. Mandatory for opening bank Letters of Credit (LC), protects landlocked trade transit, and covers **General Average** maritime sacrifices.

• **3. Auto (Motor) Insurance & Compulsory Third-Party Liability:**

- *Three Parties:* First Party (Vehicle Owner/Insured), Second Party (Insurer), Third Party (Pedestrians, other vehicle passengers/property).
- *Compulsory Third-Party Motor Insurance:* Mandatory by law in Nepal; covers bodily injury, permanent disability, death, and property damage caused to third parties (does NOT cover damage to the insured's own vehicle, which requires a Comprehensive Motor Policy).
- *Major Issues in Auto Insurance:* High accident frequency, inflated garage repair bills (moral hazard), tariff rigidities, uninsured rural vehicles, and high EV battery/spare part costs.

• **4. Agriculture Insurance & Why Crucial for Nepal:**

- Covers crops, livestock, poultry, fish, and herbs against natural and biological perils (supported by an 80% government premium subsidy in Nepal). Crucial because >60% of Nepalese depend on agriculture, terrain is climate-vulnerable, and insurance acts as collateral for agricultural bank loans.

• **5. Social Insurance (e.g., Nepal's Social Security Fund — SSF):**

- Compulsory, government-mandated programs financed by employer (20%) + employee (11%) = 31% payroll contributions covering: (i) Medical, Health & Maternity, (ii) Accident & Disability, (iii) Dependent Family Protection, and (iv) Old-Age Pension/Gratuity.

7.1 Insurance Regulation & Role of Nepal Insurance Authority (NIA / Insurance Act 2079)

• **Why Insurance is Heavily Regulated (5 Core Reasons):**

1. **Maintain Insurer Solvency:** Protect policyholders against company bankruptcy after decades of paying premiums.
2. **Compensate for Inadequate Consumer Knowledge:** Policies are complex technical contracts of adhesion.
3. **Ensure Reasonable, Adequate, and Non-Discriminatory Rates:** Prevent destructive price wars (inadequate rates) and monopoly price-gouging (excessive rates).
4. **Make Insurance Available & Promote Financial Inclusion:** Micro-insurance, rural access, and agriculture insurance.
5. **Prevent Unfair Market Practices & Enforce Prompt Claim Settlement.**

• **Major Areas of Regulation & Functions of Nepal Insurance Authority (Beema Pradhikaran under Insurance Act 2079):**

(1) **Licensing & Paid-Up Capital** (Rs 5 Billion for Life Insurers; Rs 2.5 Billion for Non-Life Insurers; Rs 750 Million for Micro-Insurers), (2) **Solvency Margin & Risk-Based Capital Reserves**, (3) **Investment Directives** (guiding where technical reserves are invested), (4) **Policy Form & Tariff Approval**, (5) **Licensing Intermediaries** (Agents, Brokers, Surveyors), and (6) **Quasi-Judicial Claim Dispute Resolution**.

7.2 Exam Master Guide: How to Score 18+/20 in Group D Case Studies (BNK 209)

In BNK 209, Group D Case Studies follow two classic patterns—be ready for both:

• **Pattern 1 — Entrepreneur / IT Consultant / Professional Case (e.g., Aarav Sharma or Bikash Koirala):**

- **Part (a) Identify Pure vs. Speculative Risks:** Always separate into: *Pure Risks* (Premature death, disability, critical illness, office fire/earthquake/theft of servers & laptops, client data breach / professional negligence liability, employee illness) vs. *Speculative Risks* (Stock market investing/trading and startup business venture profit/loss).
- **Part (b) Suitable Life Policy + Riders:** Recommend **Term Life Insurance** (highest coverage for lowest premium to cover home/car loans and children's education while cash is tied up in the startup, or "Buy Term and Invest the Difference" for a CA/investor) + 3 Riders: **(1) Critical Illness Rider, (2) Permanent Total Disability / AD&D; Rider, and (3) Waiver of Premium Rider**.
- **Part (c) Property or Professional Liability Policy:** Recommend **Professional Indemnity (Errors & Omissions / Cyber Liability) Insurance** (covers client damages + legal defense fees for employee configuration errors/data breaches) and **Standard Fire & Allied Perils + Electronic Equipment Insurance (EEI)** for office servers and laptops.
- **Part (d) Why Group Health Insurance is a Risk Management Tool:** Explains how it controls **Human Resource Risk** (reduces absenteeism, retains scarce skilled talent, lowers employee financial stress/errors, and provides predictable tax-deductible cost).

• **Pattern 2 — Quantitative Family Financial Planning Case (e.g., Shree Sharma):**

- Apply **HLV = Annual Family Share x PVIFA(r, n)**, **Needs Approach Table**, policy/rider justification, and **1-Year Term NSP = (Face Value x q_x) / (1 + i)**.